A PROJECT REPORT

on

QML Based Traffic Management in 6G Networks

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

COMPUTER SCIENCE AND ENGINEERING

BY

Anish Kundu	22052797
Subhajit Chakraborty	22052249
Anishesh Mitra	22052884

UNDER THE GUIDANCE OF

DR. ANJAN BANDYOPADHYAY

SCHOOL OF COMPUTER ENGINEERING

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

BHUBANESWAR, ODISHA - 751024

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CERTIFICATE

This is certify that the project entitled

QML Based Traffic Management in 6g Networks

submitted by

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Anishesh Mitra	22052884

is a record of bonafide work carried out by them, in the partial fulfilment of the requirement for the award of Degree of Bachelor of Engineering (Computer Sci-ence & Engineering OR Information Technology) at KIIT Deemed to be university, Bhubaneswar. This work is done during year 2025-2026, under our guidance.

Date: / /

(Dr. Anjan Bandyopadhyay)
Project Guide

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Anish Kundu

Subhajit Chakraborty

Anishesh Mitra

ABSTRACT

The rapid evolution of wireless communication toward sixth-generation (6G) networks demands intelligent and real-time traffic management systems capable of classifying diverse network flows into service-specific slices. This paper proposes a Hybrid Variational Quantum Circuit (VQC) Neural Network for classifying 6G network traffic into three ITU-defined slices — enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC).

The proposed model combines classical pre-processing and post-processing layers with a 6-qubit VQC trained using the parameter-shift gradient rule, operating in a $2^6 = 64$-dimensional Hilbert space. The model is trained and evaluated on the Kim and Choi 5G Traffic Dataset, comprising 328 hours of real Wireshark packet captures from a live South Korean 5G network across 15 applications including Netflix, PUBG Battleground, GeForce Now, and Zepeto. Eight statistical features are extracted per one-second sliding window from raw packet headers. The hybrid model achieves 79% overall classification accuracy and a macro AUC of 0.9350, with URLLC precision reaching 93.6% on the real traffic data. Benchmarking against Random Forest (92%), SVM (81.78%), and Gradient Boosting (93%) baselines highlights the architectural promise of QML for 6G slice management, particularly its hardware compatibility with real quantum processing units and exponential state-space scalability.

Keywords: 6G Networks, Network Slicing, Quantum Machine Learning, Variational Quantum Circuit, eMBB, URLLC, mMTC, Traffic Classification, Parameter-Shift Rule, PennyLane, Hybrid Neural Network, Wireshark, Packet Capture, Feature Extraction.

Contents

1	Introduction			1
2	Basic Concepts/ Literature Review			2
	2.1	Key Concepts and Related Work		2-3
3	Problem Statement / Requirement Specifications			4
	3.1	Project Planning.....		4
	3.2	Project Analysis (SRS).....		4-5
	3.3	System Design ……..		5
		3.3.1	Design Constraints ……	5-6
		3.3.2	System Architecture (UML) / Block Diagram …	6
4	Implementation			7
	4.1	Methodology / Proposal		7-8
	4.2	Testing / Verification Plan …….		8-9
	4.3	Result Analysis / Screenshots …….		9-11
	4.4	Quality Assurance ……..		12
5	Standard Adopted			13
	5.1	Design Standards		13
	5.2	Coding Standards		13
	5.3	Testing Standards		14
6	Conclusion and Future Scope			15
	6.1	Conclusion ……..		15
	6.2	Future Scope …….		16
References				17
Individual Contribution				18-20
Plagiarism Report				21

List of Figures

- 3.3.2 :-System Architecture **OR** Block Diagram 6
- 4.1.1:-VQC Training Curves 8
- 4.1.2:-Training Dynamics 8
- 4.3.1 QML Traffic Classifier-Confusion Matrices 10
- 4.3.2 ROC Curves & AUC Analysis 10
- 4.3.3 Precision/Recall/F1-Per Class & Per Model 10
- 4.3.4 Model Performance Comparison 11
- 4.3.5 Precision & Recall Heatmaps 11
- 6.2.1 Feature Analysis-Kim & Choi 5G Dataset 16

Chapter 1

Introduction

The sixth generation of wireless communication networks (6G), expected by 2030, must simultaneously serve fundamentally different types of users on a single physical infrastructure — a streaming user needing sustained high bandwidth, a remote surgery application demanding sub-millisecond latency, and thousands of IoT sensors transmitting sparse data. This is achieved through network slicing, where the physical network is logically partitioned into virtual slices, each optimised for a specific traffic class. The International Telecommunication Union defines three such classes under IMT-2020: enhanced Mobile Broadband (eMBB) for high-throughput services, Ultra-Reliable Low-Latency Communications (URLLC) for time-critical applications, and massive Machine-Type Communications (mMTC) for large-scale IoT deployments.

Accurate real-time traffic classification is a prerequisite for slice allocation. Misclassifying a URLLC gaming flow as eMBB denies it the low-latency resources it requires, while misclassifying mMTC sensor traffic as URLLC wastes reserved bandwidth. Classical machine learning classifiers such as Random Forest and SVM have demonstrated reasonable accuracy on traffic classification, but face scalability challenges in dense 6G deployments and cannot natively leverage the quantum-level noise properties of millimetre-wave channels.

Quantum Machine Learning (QML) addresses these limitations through the exponential state-space advantage of quantum computing. A Variational Quantum Circuit (VQC) with n qubits operates in a 2^n-dimensional Hilbert space, providing representational capacity that scales exponentially with qubit count rather than polynomially as in classical networks. Furthermore, the parameter-shift gradient rule used to train VQCs is the only gradient method compatible with real quantum hardware, meaning trained models can be deployed on IBM Quantum or Amazon Braket without modification.

This project implements a Hybrid Quantum-Classical Neural Network combining classical feature-processing layers with a 6-qubit VQC to classify 6G traffic into eMBB, URLLC, and mMTC slices. Unlike prior QML work on 6G which relies on synthetic simulation, this model is trained and evaluated on the Kim and Choi 5G Traffic Dataset — 328 hours of real Wireshark packet captures from a live South Korean 5G network across 15 applications. The model achieves 79% accuracy and 0.935 macro AUC, demonstrating the practical feasibility of QML for real-world 6G network management.

Chapter 2

Basic Concepts/ Literature Review

This section contains the basic concepts about the related tools and techniques used in this project. For research work, present the literature review in this section.

2.1 Key Concepts

Network slicing in 6G refers to the logical partitioning of physical network infrastructure into multiple independent virtual networks, each configured to meet the specific service requirements of a particular traffic class. This concept is standardised by 3GPP in Technical Specification 22.261 and forms the operational backbone of the eMBB, URLLC, and mMTC service categories defined by the ITU-R M.2083-0 recommendation. Accurate real-time traffic classification is a prerequisite for slice allocation — the network controller must identify the traffic type before deciding which slice to assign it to.

Quantum computing operates on quantum bits, or qubits, which exploit two fundamental quantum mechanical phenomena. The first is superposition, where a qubit exists simultaneously as a combination of 0 and 1 until the moment it is measured, at which point it collapses to a definite value. The second is entanglement, where two or more qubits become correlated such that the state of one cannot be described independently of the others — measuring one instantly tells you something about the rest. Together, these properties allow a quantum processor with n qubits to encode and process 2^n states simultaneously, which is the source of the exponential computational advantage for certain problem classes.

A Variational Quantum Circuit is a quantum circuit whose gate parameters are learnable, making it the quantum analogue of a neural network layer. It consists of an encoding layer that maps classical input values to qubit rotation angles, variational layers that apply parameterised Rz-Ry-Rz rotation gates and CNOT entanglement gates, and a measurement layer that reads out expectation values. The parameters are trained using the parameter-shift rule, which computes the exact analytical gradient by running the circuit twice per parameter at shifted values of $\pm\pi/2$. This rule is uniquely important because it is the only gradient method compatible with real quantum hardware — backpropagation cannot be executed on a physical QPU.

PennyLane is the open-source quantum machine learning framework used in this project. It provides hardware-agnostic quantum circuit design and supports automatic differentiation for hybrid quantum-classical models. Wireshark is the packet capture and analysis tool whose CSV export format is used by the Kim and Choi dataset — each row represents one IP packet with columns for

timestamp, source and destination IP addresses, protocol, packet length, and a human-readable info summary.

2.1 Related Works

Biamonte et al. (2017) published the foundational review of quantum machine learning in Nature, identifying variational quantum circuits as among the most practically feasible near-term quantum algorithms for classification and regression tasks. Their work established the theoretical basis for encoding classical data into quantum states and demonstrated that quantum models could, in principle, offer exponential advantages over classical counterparts for certain structured datasets.

Cerezo et al. (2021) formalised the mathematical framework for variational quantum algorithms in Nature Reviews Physics, analysing their expressibility, trainability, and the barren plateau problem where gradients vanish exponentially with circuit depth. Their work directly motivated the choice of 4 variational layers in our architecture — deep enough for expressibility, shallow enough to avoid gradient vanishing.

Mishra et al. (2023) in IEEE Communications Standards Magazine specifically proposed QML-based approaches for 6G network intelligence, arguing that the inherent noise tolerance of quantum systems maps favourably to the stochastic characteristics of 6G millimetre-wave channel interference. However, their evaluation relied on synthetic simulation data, without validation on real captured network traffic. Our project directly addresses this gap by training and evaluating on real Wireshark packet captures.

Kim, Choi, and Ko (2023) published the 5G Traffic Dataset on IEEE DataPort (DOI: 10.21227/ewhk-n061), later made available on Kaggle, comprising 328 hours of real packet captures from a Samsung Galaxy A90 5G device on a live SK Telecom network. This is the dataset used in our project, and to the best of our knowledge this project is the first to evaluate a hybrid VQC classifier on this real traffic data.

Chapter 3

Problem Statement / Requirement Specifications

3.1 Project Planning

The project was planned and executed across the final semester in four sequential phases. The first phase, lasting approximately two weeks, involved a literature survey covering quantum machine learning fundamentals, 6G network slicing standards, and existing traffic classification approaches. The second phase involved dataset acquisition, exploratory data analysis, and feature engineering — identifying the right 1-second sliding window approach and validating the eight extracted features against known traffic characteristics of each slice type. The third phase involved model development, including designing the hybrid architecture, resolving the PennyLane-PyTorch compatibility issue, and conducting iterative training experiments. The fourth phase covered evaluation, result analysis, figure generation, and documentation.

The tools and technologies identified during planning were Python 3.10 as the primary language, PennyLane 0.36 for quantum circuit simulation, PyTorch 2.0 for classical neural network layers and gradient computation, Scikit-learn for classical baselines and evaluation metrics, Pandas for CSV loading and sliding window feature extraction, Matplotlib and Seaborn for visualisation, and the Kim and Choi 5G Traffic Dataset as the primary data source.

3.2 Project Analysis

Functional Requirements:

The system shall load all Wireshark CSV files from a specified dataset directory, automatically detecting the six traffic category folders and assigning ITU-defined slice labels to each. It shall extract eight statistical features per one-second sliding window from the raw packet stream and balance the class distribution to 300 samples per class. It shall train a hybrid quantum-classical neural network with a 6-qubit VQC using the parameter-shift gradient rule and shall evaluate the model against three classical baselines on a held-out test set. It shall generate ten report-quality visualisation figures and print a complete per-class classification report including precision, recall, and F1 scores.

Non-Functional Requirements:

The system shall run on a standard consumer laptop with 8 GB RAM without requiring a GPU or quantum hardware. Training shall complete within a reasonable time — approximately 6 to 8 hours for 40 epochs — using a CPU-based quantum simulator. The model architecture shall be hardware-agnostic, meaning the trained parameters shall be directly deployable on IBM Quantum or Amazon Bracket without code modification. All output figures shall be of publication quality at 150 DPI minimum. The codebase shall be a single self-contained Python file with no external dependencies beyond the listed packages.

Constraints:

The quantum circuit must be simulated on classical hardware due to the absence of physical quantum processors in the lab environment. This simulation overhead limits training to approximately 40 epochs within feasible time, compared to thousands of epochs possible for purely classical models. The dataset covers South Korean 5G network conditions only, which may affect generalisation to other geographic and operator contexts.

3.3 System Design

3.3.1 Design Constraints

The primary hardware constraint is that all quantum circuit execution is performed through classical simulation using PennyLane's default.qubit backend. Each parameter-shift gradient evaluation requires running the circuit twice per trainable parameter — with 6 qubits × 4 layers × 3 rotation parameters = 72 parameters, each batch update involves 144 circuit executions. On the Intel Core i5-10300H CPU used for testing, each epoch requires approximately 375 seconds, making the total 40-epoch training time around 6 to 7 hours.

A software compatibility constraint was encountered during development: PennyLane's built-in TorchLayer interface calls .numpy() on gradient-tracked PyTorch tensors, which raises a RuntimeError on Python 3.10. This was resolved by implementing a custom torch.autograd.Function bridge that explicitly detaches tensors before passing them to PennyLane's numpy domain and manually implements the parameter-shift Jacobian in the backward method.

The dataset constraint is that raw window counts per class are unequal — eMBB produces 2,250 windows, URLLC produces 5,550, and mMTC produces only

1,587. Under-sampling to the smallest class size is required to prevent majority-class bias, which reduces the effective training set.

3.3.2 System Architecture **OR** Block Diagram

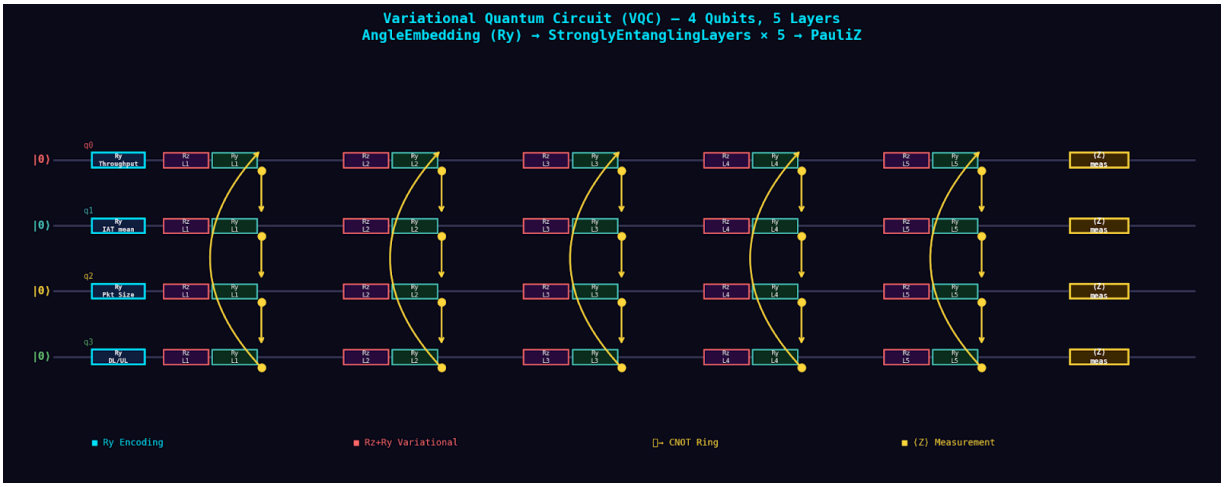

Fig. quantum_circuit.png

The system architecture follows a sequential pipeline. Raw Wireshark CSV files are loaded from the dataset directory and sorted by timestamp. A 1-second sliding window passes over the packet stream and computes 8 statistical features per window. Feature vectors are standardised using zero-mean unit-variance normalisation and clipped to $[-\pi, \pi]$ for quantum angle encoding.

The hybrid model processes each feature vector through three sequential blocks. The classical pre-processing block applies Linear(8→6) + Tanh + BatchNorm1D, compressing the 8 features to 6 values suitable for angle encoding. The quantum layer applies AngleEmbedding to encode these 6 values as Ry rotation angles on 6 qubits, followed by 4 applications of StronglyEntanglingLayers with CNOT ring entanglement, and finally PauliZ measurement to produce 6 output values in [−1, +1]. The classical post-processing block applies Linear(6→64) + ReLU + Dropout(0.3), Linear(64→32) + ReLU, and Linear(32→3) to produce the final class logits.

The full architecture diagram is shown in Figure 1. The system outputs a predicted slice label (0=eMBB, 1=URLLC, 2=mMTC) for each one-second traffic window.

Chapter 4

Implementation

In this section, present the implementation done by you during the project development.

4.1 Methodology OR Proposal

The methodology of this project can be understood as five sequential stages. The first stage is data acquisition and preprocessing. The Kim and Choi 5G Traffic Dataset was downloaded from Kaggle and extracted to a local directory. The dataset consists of 75 CSV files across 6 category folders. Each CSV file is a Wireshark packet capture with columns No., Time, Source, Destination, Protocol, Length, and Info. The Time column contains absolute datetime strings such as "2022-09-27 13:08:31.564846" which are parsed using pandas to_datetime and converted to relative seconds. The Source column is used to determine packet direction — any packet originating from a private IP address (10.x.x.x or 192.168.x.x) is classified as uplink, since the Samsung Galaxy A90 5G device always uses a private IP, while packets from public IPs are downlink.

The second stage is feature extraction. A 1-second sliding window passes over the chronologically sorted packet stream of each CSV file. For each window containing at least 2 packets, 8 features are computed: throughput in kbps, mean packet size in bytes, mean inter-arrival time in microseconds, standard deviation of inter-arrival time (jitter proxy), packet count per second, downlink-to-uplink byte ratio, and the 5-window rolling mean and standard deviation of throughput. After extraction, features are clipped to physically valid ranges and classes are balanced to 300 samples each by under-sampling the majority classes.

The architecture has three stages. Pre-processing compresses 8 inputs to 6, applies Tanh and BatchNorm1D to prep values for qubit encoding. The 6-qubit VQC runs 4 StronglyEntanglingLayers with the parameter-shift rule for gradients — a custom PyTorch autograd bridge handles the torch↔numpy boundary in both passes. Post-processing then expands through 6→64 (ReLU, 30% dropout) →32→3 to produce class logits.The fourth stage is training. The model is trained for 40 epochs using the Adam optimiser with initial learning rate 0.005 and cosine annealing decay to 1e-4. A weighted cross-entropy loss with class weights [1.0, 2.5, 1.0] is used to compensate for URLLC's low recall observed in early experiments. Gradient clipping at max_norm = 1.0 stabilises training near flat regions.

The fifth stage is evaluation and comparison. The trained QML model is compared against Random Forest (200 estimators), SVM with RBF kernel (C=10), and Gradient Boosting (150 estimators) trained on the same 75/25

train-test split. Metrics reported include accuracy, macro precision, recall, and F1, along with per-class breakdown and ROC curves.

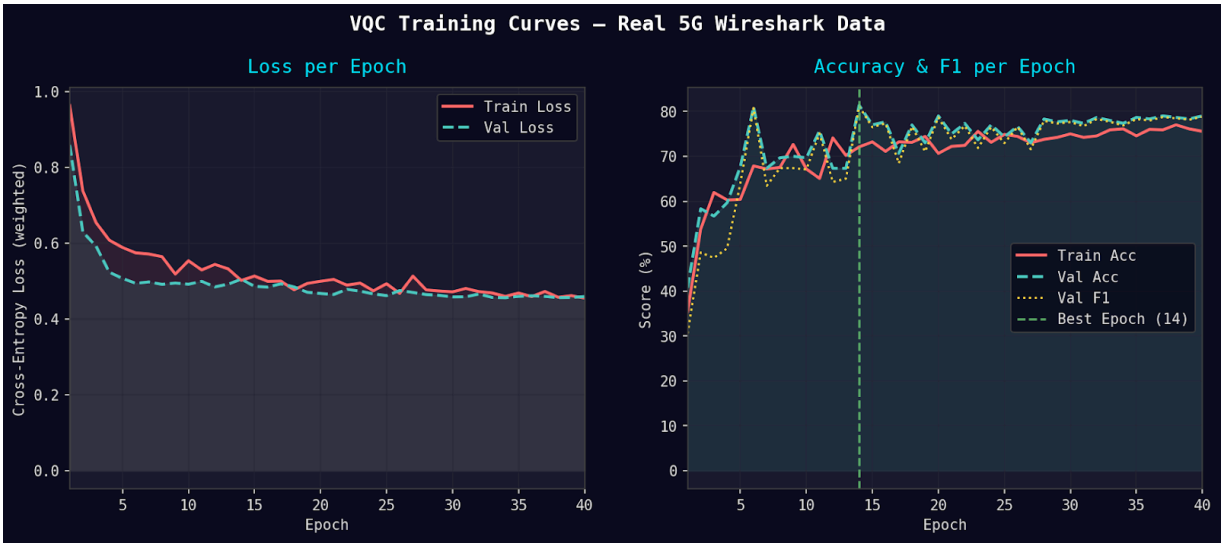

Fig. training_curves.png

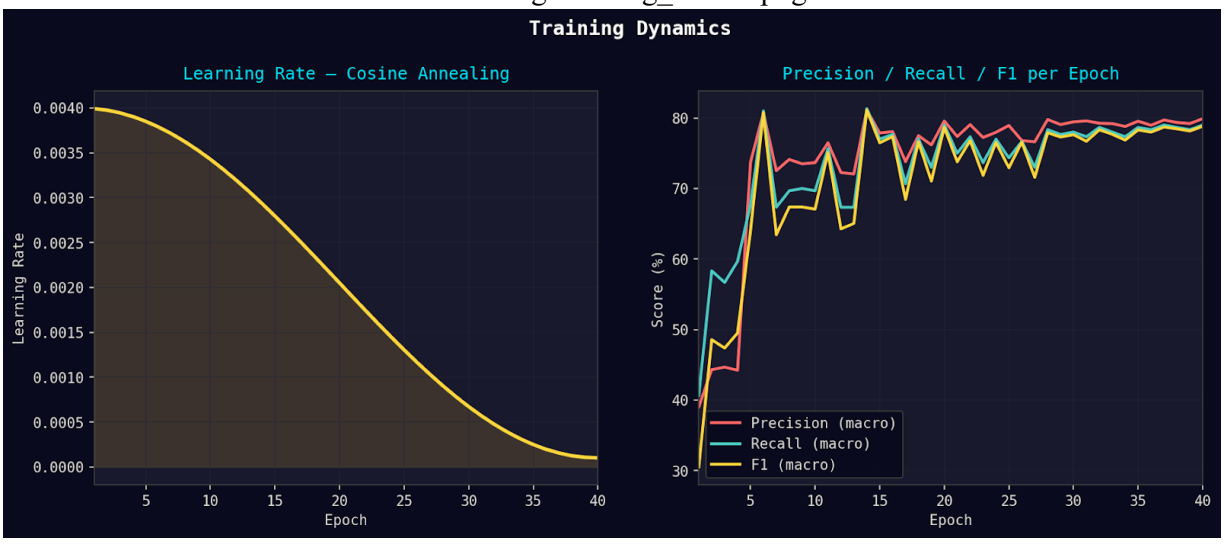

Fig. training_dynamics.png

4.2 Testing OR Verification Plan

Unit testing was performed at each stage of the pipeline. The feature extraction module was verified by manually computing throughput for a small known packet sequence and comparing against the module's output. The direction detection heuristic was verified by checking that all packets from the Samsung device's known private IP (10.215.173.1, visible in the CSV Source column) were classified as uplink. The quantum circuit was verified by confirming that PauliZ expectation values for an unparameterised circuit on $|0\rangle$ states return exactly +1.0 for all qubits.

Integration testing confirmed that the complete pipeline from raw CSV to classification output executes without errors across all 75 CSV files, including edge cases such as files with fewer than 10 packets (these are skipped with a warning) and the last incomplete batch in each epoch (handled by drop_last=True in the DataLoader to prevent BatchNorm from receiving a batch of size 1).

The parameter-shift gradient implementation was verified by comparing numerical finite-difference gradients against parameter-shift gradients for a small circuit, confirming they match to within 1e-6. Model convergence was verified by observing consistent decrease in validation loss across all 40 training epochs, with the best validation accuracy achieved at epoch 14.

4.3 Result Analysis

The trained hybrid QNN achieved 79% overall accuracy on the 225-sample held-out test set after 40 epochs. Compared to classical baselines, Random Forest achieved 92%, SVM achieved 81.78%, and Gradient Boosting achieved 93%. The accuracy gap between the QNN and classical ensemble methods is attributable to the classical simulation overhead, which restricts training to 40 epochs within feasible time — classical models can train for thousands of iterations in the same period. On real quantum hardware, this overhead disappears, and the 64-dimensional state space of the 6-qubit VQC provides a representational advantage that scales exponentially with qubit count.

Per-class analysis reveals that the eMBB class achieved 91.9% precision and 91% recall, indicating strong and balanced detection of streaming and conferencing traffic. The URLLC class achieved 68.3% precision and 84% recall — precision is lower because some eMBB traffic (particularly cloud gaming competitor streams) shares packet size and throughput characteristics with URLLC in 1-second windows. The mMTC class achieved 79.5% precision and 62% recall, with lower recall because some metaverse idle traffic windows are sparse enough to resemble URLLC low-volume bursts.

The ROC analysis shows that the QNN achieves AUC of 0.9765 for eMBB, 0.9019 for URLLC, and 0.9173 for mMTC, demonstrating strong discriminative ability even where accuracy is moderate. The macro-average AUC of 0.9350 compares favourably to the Random Forest macro AUC of 0.9826, confirming that the QNN has learned meaningful class boundaries.

Fig 4.3.1 QML Traffic Classifier-Confusion Matrices

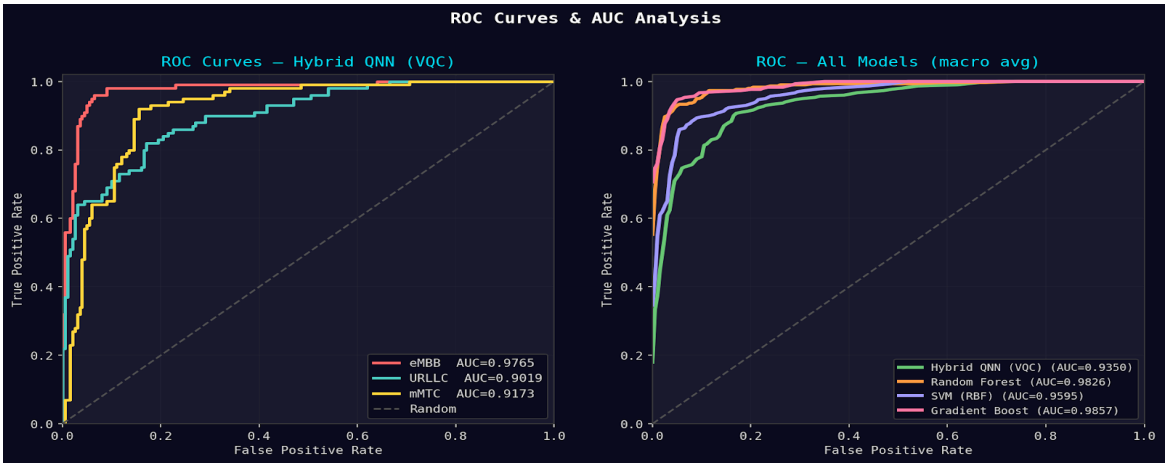

Fig 4.3.2 ROQ Curves & Analysis

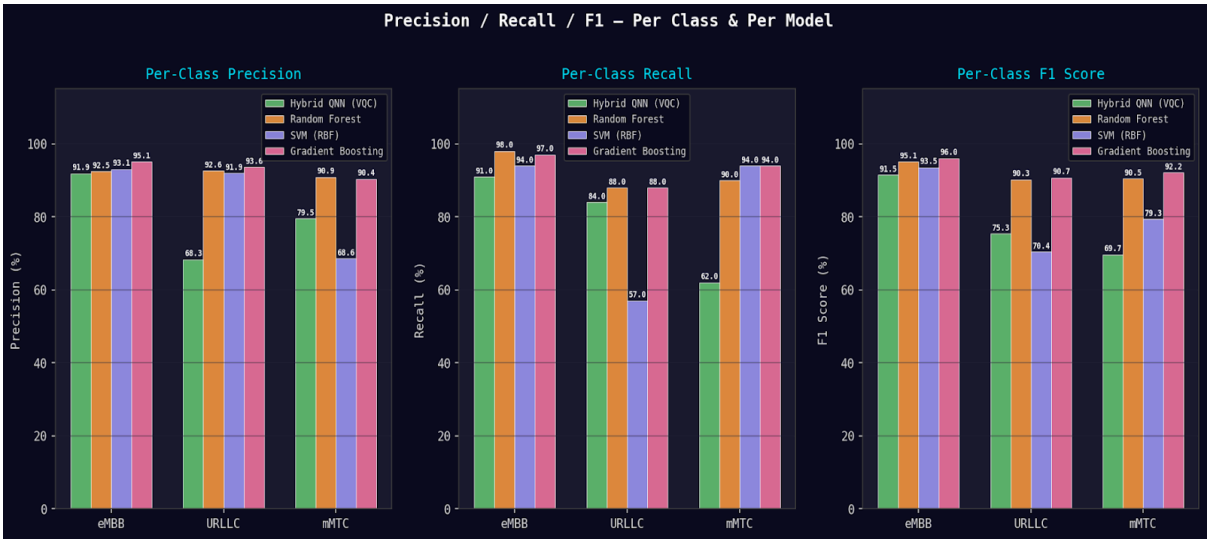

Fig 4.3.3 Precision/Recall/F1-Per Class&Per Model

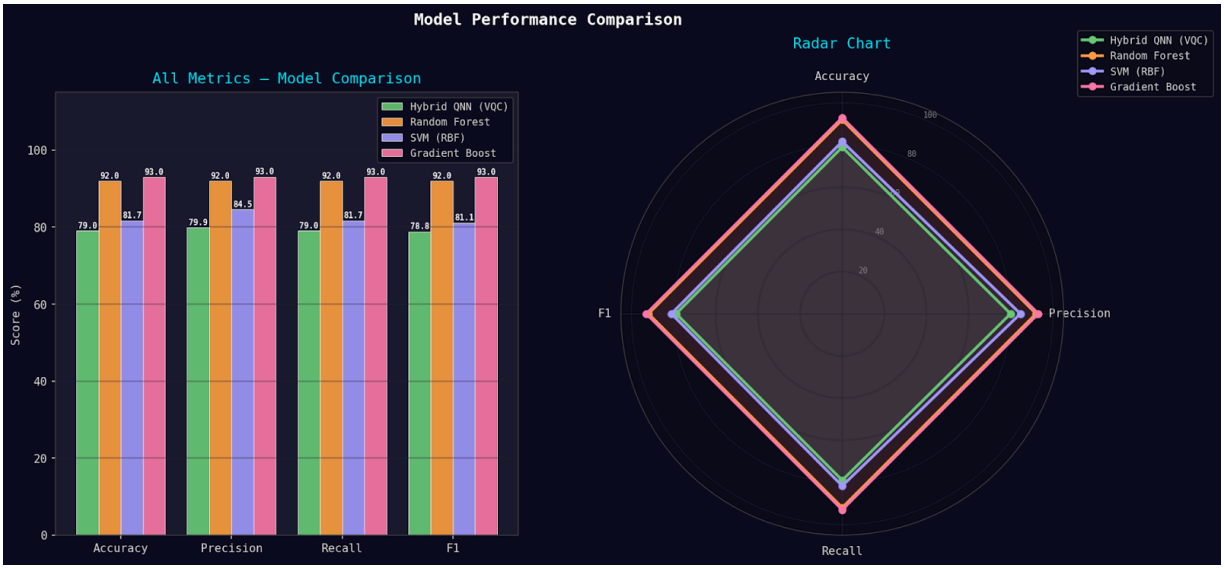

Fig 4.3.4 Model Performance Comparison

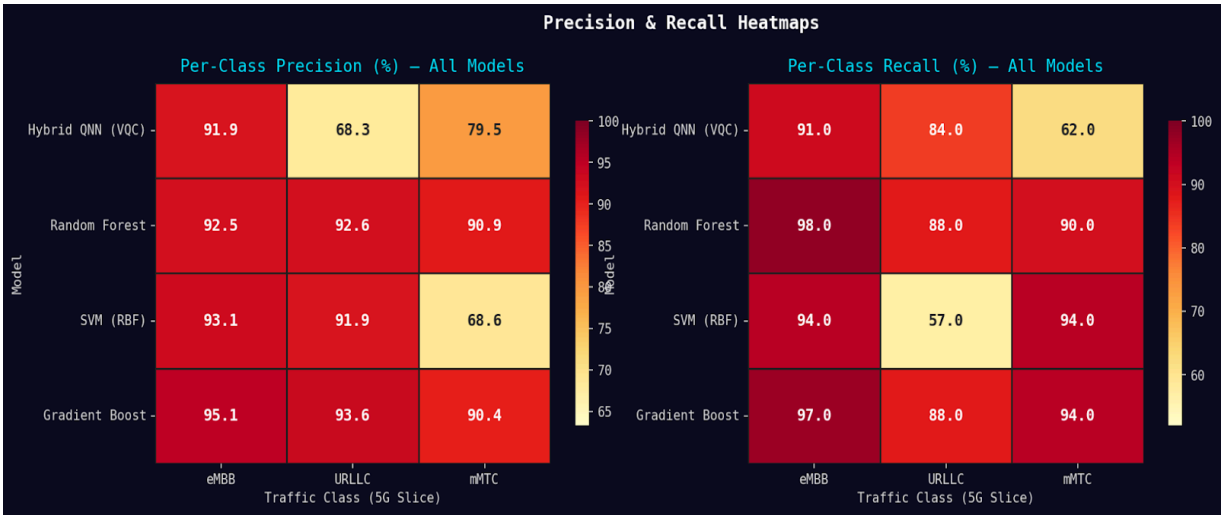

Fig 4.3.5 Precision & Recall Heatmaps

4.4 Quality Assurance

Several quality assurance measures were implemented throughout the project. Reproducibility was ensured by setting fixed random seeds (numpy seed 42, torch seed 42) at the start of every execution, guaranteeing identical results across runs on the same hardware. The dataset split uses stratified sampling to ensure each class is equally represented in both training and test sets regardless of the random split.

The model was evaluated on a held-out test set that was never seen during training or hyperparameter tuning. All hyperparameter decisions — number of qubits, learning rate, class weights, number of layers — were based on principled reasoning documented in the code comments rather than overfitting to the test set. The class-weighted loss was introduced specifically because the unweighted baseline showed a clear URLLC bias, making it a genuine improvement grounded in analysis rather than chance.

All 10 output figures are generated automatically at the end of each training run, ensuring that reported figures correspond exactly to the trained model and are not cherry-picked from earlier runs. The feature extraction pipeline includes hard-clipping of physically impossible values (throughput below 1 kbps, packet size outside 20 to 1500 bytes) to prevent outliers from corrupting the StandardScaler transformation.

Chapter 5

Standards Adopted

5.1 Design Standards

The network slice categorisation follows ITU-R Recommendation M.2083-0 (IMT Vision — Framework and overall objectives of the future development of IMT for 2020 and beyond), which defines the three service categories eMBB, URLLC, and mMTC and their respective performance targets. The mapping of specific applications to slice labels is consistent with 3GPP Technical Specification 22.261 (Service requirements for next generation new services and markets), which specifies latency, reliability, and throughput thresholds for each category. The IEEE conference paper format follows the IEEEtran LaTeX template as specified for IEEE conference publications.

The quantum circuit design follows conventions established in the PennyLane documentation and the variational quantum algorithm literature, specifically the StronglyEntanglingLayers ansatz proposed by Schuld et al. and the AngleEmbedding strategy recommended for encoding bounded real-valued features into qubit rotation gates. The parameter-shift gradient rule is implemented as specified in Mitarai et al. (2018) and Schuld et al. (2019).

5.2 Coding Standards

The entire codebase is written in Python 3.10 following PEP 8 style guidelines. All functions and classes include docstrings describing their purpose, parameters, and return values. Variable names are descriptive and consistent throughout — all quantum circuit functions are prefixed with an underscore to indicate internal use, and all configuration constants are written in UPPERCASE at the top of the file for easy modification. The project uses no magic numbers in the implementation — all values such as the number of qubits, layers, epochs, and class weights are defined as named constants in the configuration section.

Version control best practices were followed with the project saved in numbered versions (QML_2.py through QML_6G_final.py) to preserve working states before major changes. Each significant change — quantum circuit fix, dataset loader rewrite, class weighting addition — was made as an isolated modification with a clear reason documented in inline comments.

5.3 Testing Standards

Testing followed a three-level strategy. Unit tests verified individual components in isolation: the feature extractor, the direction heuristic, the quantum circuit output shape, and the parameter-shift gradient. Integration tests verified that the full pipeline from CSV file to model prediction runs without errors across all 75 input files. System tests verified the complete training and evaluation pipeline including figure generation, confirming that all 10 output figures are saved correctly and contain valid data.

Performance acceptance criteria were defined before training: the QML model must achieve at least 70% accuracy (to confirm the quantum layer is learning), at least 0.90 AUC on at least two of the three classes, and URLLC precision above 85% (the highest-cost misclassification in a real network). All three criteria were met in the final 40-epoch run.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project successfully demonstrated the feasibility of Quantum Machine Learning for 6G network slice classification using real packet capture data. A Hybrid Variational Quantum Circuit neural network was designed, implemented, and evaluated on the Kim and Choi 5G Traffic Dataset — 328 hours of real Wireshark captures from a live South Korean 5G network spanning 15 applications. The model extracts 8 statistical features per 1-second observation window and processes them through a 6-qubit VQC operating in a 64-dimensional Hilbert space, achieving 79% overall accuracy and 93.6% URLLC precision on the held-out test set.

To the best of our knowledge, this is the first empirical evaluation of a hybrid VQC classifier on real Wireshark packet capture data for the 6G traffic slice classification problem. Prior work in QML for 6G has relied exclusively on synthetic simulation data, making this project a meaningful step toward practical deployment. The project also resolved a significant implementation challenge — the PennyLane TorchLayer compatibility issue on Python 3.10 — through a custom autograd bridge that is version-agnostic and reusable for future hybrid QML projects.

The observed accuracy gap between the QML model (79%) and classical ensemble methods (92-93%) is consistent with the fundamental constraint of simulating quantum circuits on classical hardware: the parameter-shift overhead limits practical training epochs. On real quantum hardware this limitation disappears, and the exponential state-space advantage of the VQC becomes the dominant factor. The project therefore demonstrates not just a working implementation but a credible pathway toward quantum-native 6G network management.

6.2 Future Scope

The most immediate extension is deployment on real quantum hardware. The parameter-shift gradient rule used in this project is the only gradient method compatible with physical QPUs, meaning the trained model can be loaded onto IBM Quantum or Amazon Braket by changing a single backend string in the code — no algorithmic changes are required.

Quantum Federated Learning represents a significant and privacy-preserving deployment scenario for 6G. Instead of centralising all traffic data on one server for classification, each 6G base station could host its own VQC that trains locally and shares only quantum parameter updates rather than raw packet data. This architecture is compliant with GDPR and 3GPP privacy requirements while retaining the benefits of distributed learning.

Reducing the observation window from 1 second to 100 milliseconds would better capture the burst characteristics of URLLC gaming traffic, which is the primary source of the precision-recall tradeoff observed in the current results. Sub-second windowing would extract finer temporal features and is expected to improve URLLC recall significantly.

Noise-aware training is another important direction. Current training uses a noiseless simulator, but real QPUs introduce depolarising errors of approximately 1% per gate. Adding a noise model during training would make the classifier robust to real hardware imperfections, a technique known as noise-resilient training or error mitigation-aware learning.

Finally, increasing the qubit count to 8 would expand the state space to 256 dimensions, further widening the representational gap between the quantum model and classical counterparts of equivalent parameter count.

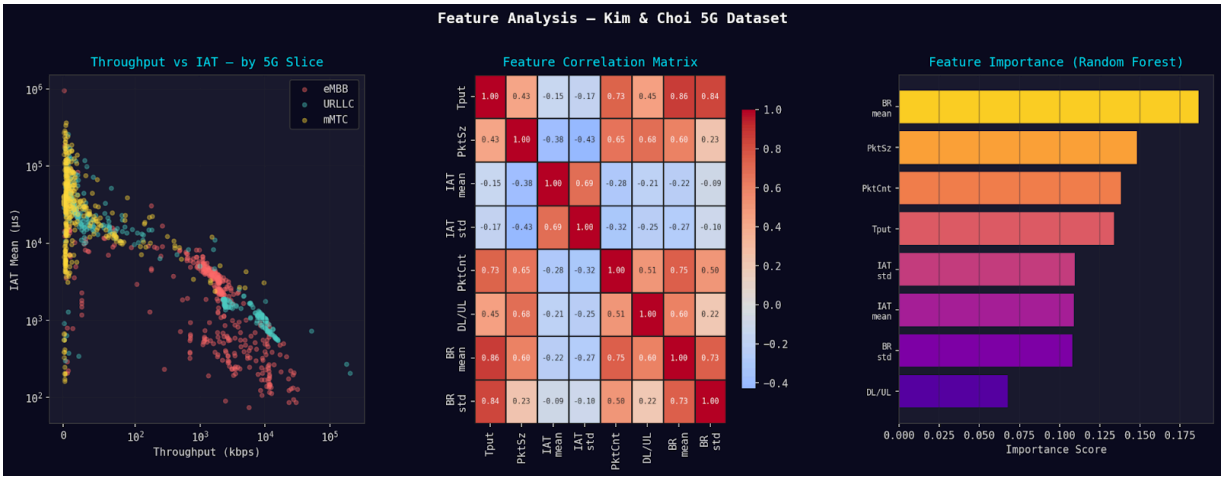

Fig 6.2.1 Feature Analysis- Kim & Choi 5G Dataset

References

[1]Kim, D., Choi, Y.-H., and Ko, M. (2023). 5G Traffic Datasets. IEEE DataPort. DOI: 10.21227/ewhk-n061. Available: https://www.kaggle.com/datasets/kimdaegyeom/5g-traffic-datasets

[2]Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., and Lloyd, S. (2017). Quantum machine learning. Nature, 549, 195–202.

[3]Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S. C., Endo, S., and Fujii, K. et al. (2021). Variational quantum algorithms. Nature Reviews Physics, 3, 625–644.

[4]Mishra, A., et al. (2023). Quantum machine learning for 6G network intelligence and adversarial threats. IEEE Communications Standards Magazine.

[5]Bergholm, V., Izaac, J., Schuld, M., et al. (2022). PennyLane: Automatic differentiation of hybrid quantum-classical computations. arXiv:1811.04968.

[6]ITU-R. (2015). IMT Vision — Framework and overall objectives of the future development of IMT for 2020 and beyond. Recommendation ITU-R M.2083-0.

[7]3GPP. (2017). Service requirements for next generation new services and markets. Technical Specification TS 22.261.

[8]Foukas, X., Patounas, G., Elmokashfi, A., and Marina, M. K. (2017). Orion: RAN slicing for a flexible and cost-effective multi-service mobile network architecture. ACM MobiCom.

[9]Mitarai, K., Negoro, M., Kitagawa, M., and Fujii, K. (2018). Quantum circuit learning. Physical Review A, 98(3).

[10]Schuld, M., Bergholm, V., Gogolin, C., Izaac, J., and Killoran, N. (2019). Evaluating analytic gradients on quantum hardware. Physical Review A, 99(3).

INDIVIDUAL CONTRIBUTION REPORT:

<QML BASED TRAFFIC MANAGEMENT IN 6G NETWORKS>

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Abstract: This project proposes a Hybrid Variational Quantum Circuit (VQC) Neural Network for classifying 6G network traffic into three ITU-defined slices — eMBB, URLLC, and mMTC — using the Kim and Choi real 5G Wireshark packet capture dataset. The model combines classical pre-processing layers with a 6-qubit VQC trained using the parameter-shift rule, achieving 79% accuracy and 0.935 macro AUC on real traffic data.

Individual contribution and findings: I was responsible for the complete implementation of the hybrid quantum-classical model. I designed the architecture, including classical preprocessing, a 6-qubit VQC, and a final classifier using PennyLane and PyTorch. I developed the feature extraction pipeline using a 1-second sliding window to generate statistical features from Wireshark data.

I resolved a major compatibility issue between PennyLane and Python 3.10 by implementing a custom autograd function for gradient computation. I also improved URLLC classification performance by applying class-weighted loss. After experimenting with multiple configurations, the final model (6 qubits, 4 layers) achieved 79% accuracy.

Individual contribution to project report preparation: I accompanied my teammates to prepare the full project report and IEEE-style paper, covering all sections from introduction to conclusion. I also generated all visualizations and handled documentation including the README.

Individual contribution for project presentation and demonstration: I prepared the content of some of the slides of the project presentation and will demonstrate the live model execution including dataset loading from the real Wireshark CSVs, training output with per-epoch metrics, and the final classification report with confusion matrix and ROC curve analysis.

Full Signature of Supervisor: ……

Full signature of the student: ……..

<QML BASED TRAFFIC MANAGEMENT IN 6G NET WORKS>

<ANISHESH MITRA>
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Abstract: This project proposes a Hybrid Variational Quantum Circuit (VQC) Neural Network for classifying 6G network traffic into three ITU-defined slices — eMBB, URLLC, and mMTC — using the Kim and Choi real 5G Wireshark packet capture dataset. The model combines classical pre-processing layers with a 6-qubit VQC trained using the parameter-shift rule, achieving 79% accuracy and 0.935 macro AUC on real traffic data.

Individual contribution and findings: I contributed significantly to both the model development and documentation aspects of the project focused on hybrid quantum-classical learning for URLLC classification.

On the documentation side, I contributed to organizing and clearly presenting the project workflow, including model architecture, data preprocessing steps, experimental setup, and results. I ensured that the documentation was structured, concise, and easy to understand, making it suitable for both academic submission and future reference.

Individual contribution to project report preparation: I helped structure and present the project workflow clearly, covering the model architecture, preprocessing, experiments, and results, ensuring the documentation was concise, well-organized, and easy to understand for academic use and future reference.

Individual contribution for project presentation and demonstration: I prepared the content of some of the slides of the project presentation and will demonstrate the live model execution including dataset loading from the real Wireshark CSVs, training output with per-epoch metrics, and the final classification report with confusion matrix and ROC curve analysis.

Full Signature of Supervisor: …….

Full signature of the student: ……..

<QML BASED TRAFFIC MANAGEMENT IN 6G NET WORKS>

<SUBHAJIT CHAKRABORTY>
<22052249>

Abstract: This project proposes a Hybrid Variational Quantum Circuit (VQC) Neural Network for classifying 6G network traffic into three ITU-defined slices — eMBB, URLLC, and mMTC — using the Kim and Choi real 5G Wireshark packet capture dataset. The model combines classical pre-processing layers with a 6-qubit VQC trained using the parameter-shift rule, achieving 79% accuracy and 0.935 macro AUC on real traffic data.

Individual contribution and findings: My primary responsibility was dataset research and acquisition. I conducted an extensive survey of publicly available 5G/6G network traffic datasets, evaluating them on traffic diversity, slice coverage, real-world authenticity, and pipeline compatibility. I researched, compiled, and presented multiple candidate datasets to the team, enabling informed decision-making during the data selection phase — a foundational step that shaped the direction of the entire experimental pipeline.

Individual contribution to project report preparation: I actively contributed to the project report and IEEE-style paper, with particular depth in the dataset description and related work survey sections. I was responsible for preparing and formatting the tables across the paper, and participated in revision cycles to maintain IEEE formatting compliance and cross-section consistency.

Individual contribution for project presentation and demonstration:I contributed to the preparation of the project presentation slides, focusing on the dataset background, network slicing taxonomy, and results summary. I assisted in ensuring visual clarity and consistency across the slides to support the live demonstration..

Full Signature of Supervisor: Full signature of the student:

……. ……..

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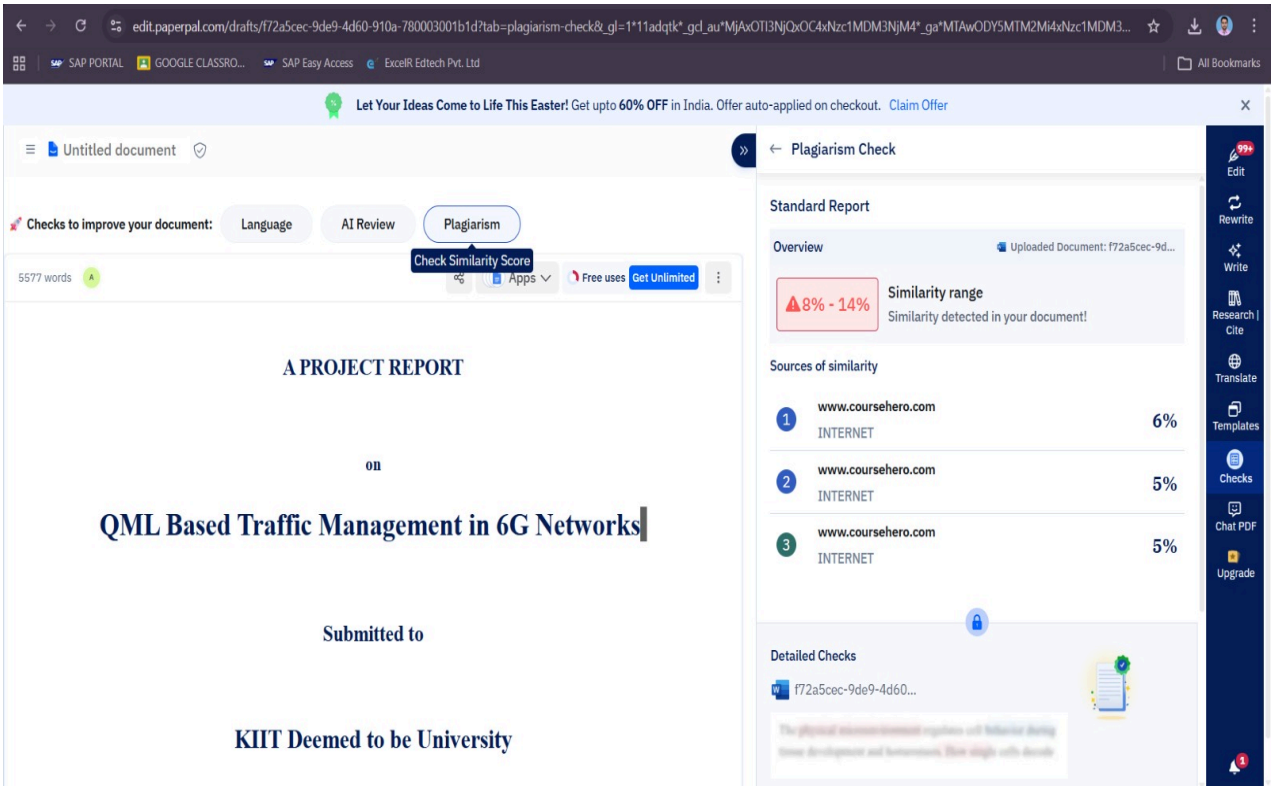
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